

Problem 1 – Question 4

State preparation

4(a): Uniform superposition

Apply a Hadamard gate to every qubit:

$$|0\rangle^{\otimes n} \xrightarrow{H^{\otimes n}} \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle.$$

The circuit has size n , depth 1, and uses no ancillary qubits. Both the size and depth are optimal: every output qubit must be changed from the fixed state $|0\rangle$, and all n one-qubit gates can run in parallel.

4(b): Unequal-amplitude GHZ state

Assume $|\alpha| \leq 1$, and choose

$$\gamma = 2 \operatorname{atan2} \left(\sqrt{1 - \alpha^2}, \alpha \right).$$

Then

$$R_Y(\gamma)|0\rangle = \alpha|0\rangle + \sqrt{1 - \alpha^2}|1\rangle.$$

Apply $R_Y(\gamma)$ to q_1 , followed by a fanout of CNOTs whose control information originates at q_1 :

$$\left(\alpha|0\rangle + \sqrt{1 - \alpha^2}|1\rangle \right) |0\rangle^{\otimes(n-1)} \mapsto \alpha|0^n\rangle + \sqrt{1 - \alpha^2}|1^n\rangle.$$

With unrestricted connectivity, use a balanced doubling tree: in layer r , every already-entangled qubit controls a fresh qubit. This requires $n - 1$ CNOTs and depth $\lceil \log_2 n \rceil$, after the initial one-qubit rotation. Hence

$$\text{size} = n, \quad \text{depth} = 1 + \lceil \log_2 n \rceil, \quad \text{ancillas} = 0.$$

On a nearest-neighbor line, the simple CNOT chain has the same size and depth n .

4(c): Four-qubit one-excitation state

Use qubit order $q_1 q_2 q_3 q_4$. Define an excitation-preserving Givens rotation by

$$\begin{aligned} G_{jk}(\varphi)|01\rangle_{jk} &= \sin \varphi |10\rangle_{jk} + \cos \varphi |01\rangle_{jk}, \\ G_{jk}(\varphi)|10\rangle_{jk} &= \cos \varphi |10\rangle_{jk} - \sin \varphi |01\rangle_{jk}, \end{aligned}$$

and let it act as the identity on $|00\rangle$ and $|11\rangle$. Starting from $|0000\rangle$, use the chronological sequence

$$X_{q_4}, \quad G_{34}(\varphi_{34}), \quad G_{23}(\varphi_{23}), \quad G_{12}(\varphi_{12}),$$

where

$$\cos \varphi_{34} = \frac{3}{10}, \quad \cos \varphi_{23} = \frac{4}{\sqrt{91}}, \quad \varphi_{12} = \frac{\pi}{4}.$$

The amplitudes evolve as

$$\begin{aligned}
|0001\rangle &\mapsto \frac{\sqrt{91}}{10}|0010\rangle + \frac{3}{10}|0001\rangle \\
&\mapsto \frac{\sqrt{3}}{2}|0100\rangle + \frac{2}{5}|0010\rangle + \frac{3}{10}|0001\rangle \\
&\mapsto \frac{\sqrt{6}}{4}|1000\rangle + \frac{\sqrt{6}}{4}|0100\rangle + \frac{2}{5}|0010\rangle + \frac{3}{10}|0001\rangle.
\end{aligned}$$

This is the requested state. It uses one X gate and three two-qubit Givens rotations, no ancillas, and depth 4. A Givens rotation can be compiled into two CNOTs and single-qubit R_Y pulses if CNOT is the native entangling gate.

4(d): Optional part retained as requested

Let the eight strings multiplying α be the affine code

$$\mathcal{A} = a_0 + \text{span}_{\mathbb{F}_2}\{g_1, g_2, g_3\},$$

where

$$\begin{aligned}
a_0 &= 1000000, \\
g_1 &= 1010101, \quad g_2 = 0110011, \quad g_3 = 0001111.
\end{aligned}$$

The eight strings multiplying β are exactly the bitwise complements of the strings in $\mathcal{A}$. Thus they form $\mathcal{A} + 1111111$. The state is normalized when

$$8(\alpha^2 + \beta^2) = 1.$$

Choose δ such that

$$\cos \frac{\delta}{2} = \sqrt{8}\alpha, \quad \sin \frac{\delta}{2} = \sqrt{8}\beta.$$

Initialize seven qubits and prepare four logical bits

$$u, v, w \text{ uniform in } \{0, 1\}, \quad s \text{ with amplitudes } \sqrt{8}\alpha, \sqrt{8}\beta,$$

by applying

$$H_{q_1}, \quad H_{q_2}, \quad R_Y(\delta)_{q_3}, \quad H_{q_4}.$$

Initially $q_1 = u, q_2 = v, q_3 = s, q_4 = w$, while $q_5 = q_6 = q_7 = 0$.

Apply the following compact linear encoder in chronological order:

$$\begin{aligned}
&\text{CNOT}_{1 \rightarrow 2}, \text{CNOT}_{1 \rightarrow 5}, \text{CNOT}_{1 \rightarrow 6}, \text{CNOT}_{2 \rightarrow 7}, \text{CNOT}_{3 \rightarrow 1}, \text{CNOT}_{1 \rightarrow 2}, \\
&\text{CNOT}_{3 \rightarrow 4}, \text{CNOT}_{7 \rightarrow 3}, \text{CNOT}_{4 \rightarrow 7}, \text{CNOT}_{7 \rightarrow 6}, \text{CNOT}_{4 \rightarrow 5}.
\end{aligned}$$

Tracking the logical bits over $\mathbb{F}_2$, the seven registers become

$$\begin{aligned}
q_1 &= u + s, & q_2 &= v + s, & q_3 &= u + v + s, \\
q_4 &= w + s, & q_5 &= u + w + s, & q_6 &= v + w + s, \\
q_7 &= u + v + w + s.
\end{aligned}$$

Finally apply X_{q_1} . The output bit string is therefore

$$a_0 + ug_1 + vg_2 + wg_3 + s(1111111).$$

Since $H^{\otimes 3}$ supplies the factor $1/\sqrt{8}$, every word with $s = 0$ has amplitude α , and every word with $s = 1$ has amplitude β , exactly as required.

This explicit encoder uses no extra ancillas. Counting the displayed primitive gates gives 3 Hadamards, 1 R_Y , 11 CNOTs, and 1 X , for total size 16. Disjoint CNOTs may be parallelized.